

Mikhail Razakov

Compiler / Runtime Engineer

Since 2024, I have been building Wist2, a public modular .NET platform for embedded DSLs and runtimes: lexer, parser and AST, bytecode/AIR, interpreter and CIL backend. I am currently completing an MCST engineering internship on the LLVM compiler-development track. Beyond compiler work, I have designed and operated .NET services with payments, migrations and recovery paths, and I have taught programming since 2021.

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Moscow / remote · HSE University, Software Engineering 2026-2030 · Available Sep 2026: internship / part-time · MCST Jul-Aug 2026

2021 — present programming teaching and project reviews	2024 — present Wist2 architecture, implementation, tests and public documentation	2026 MCST internship on the LLVM/compiler-engineering track	2023-2026 client backend, VPN service and LMS
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EXPERIENCE

Jul-Aug 2026	Engineering internship: algorithms and compiler infrastructure MCST · LLVM compiler-development track · remote	- Implementing C++23 graph components over a shared model: reverse postorder, cycle detection, shortest paths, maximum flow and strongly connected components. - Maintaining CMake builds, separate executables and reproducible input/output test harnesses. - Using strict compiler warnings, AddressSanitizer, UndefinedBehaviorSanitizer and static analysis while controlling module dependencies.
2024 — present	Creator and primary developer of Wist2 Open compiler/runtime project · .NET	- Designed the modular architecture: dialect definitions, lexer, parser and AST, bytecode/AIR, optimizers, interpreter and CIL generation through DynamicMethod. - Developing the public API and NuGet packages, module and backend-capability contracts, diagnostics, documentation and integration examples. - Maintaining regression tests and semantic-parity checks between interpreted and compiled execution; using BenchmarkDotNet without generalising beyond measured scenarios. - Exploring callable-first SSA as an optional IR layer while keeping language semantics independent from one optimisation representation.
2026	Product .NET backend development and operations VpnMediator / Razaltush VPN · CompilationLabLMS	- Designed subscriptions, access rights, payment operations, balance ledgers, verified webhooks, migrations and background recovery workers. - Worked on diagnostics, backup and restore, systemd/nginx, versioned releases and rollback after failed health checks. - Supported user and operator flows after launch, including retries and partial failures.

ADDITIONAL EXPERIENCE

- 2023**
Client licensing backend
Private small-business project · ASP.NET Core / OpenAPI
 - Implemented key issuance and validation, activation states and client-application integration.
 - Documented the contract through OpenAPI/Swagger and an explicit error model; client code and data are private.
- 2021 — present**
Programming tutor and mentor
About 50 students · C/C++, C#, Python, Unity and algorithms
 - Teaching algorithms, program structure and engineering trade-offs; reviewing code and diagnosing knowledge gaps.
 - Helping students bring projects to a working demonstration or defence rather than solving isolated tasks only.

SELECTED PROJECTS

COMPILER/RUNTIME PLATFORM

Wist2

- Pipeline: text → lexer/parser/AST → bytecode/AIR → optimizers → interpreter or CIL

[GitHub](#) · [Documentation](#)

PROGRAM ANALYSIS

PS-form Memory Dependence Analyzer

C17 solver that determines whether parametric memory accesses may overlap across loop iterations.

- Ranked 1st of 49 submissions; 5.0/5.0 and 104/104 tests.

[GitHub](#)

MCST ENGINEERING INTERNSHIP

Algorithmic Components and C++23 Infrastructure

C++23 algorithms over a shared graph model: RPO, cycle detection, Dijkstra, Dinic and Tarjan SCC.

- CMake, separate executables and a reproducible input/output test harness.

[GitHub](#)

SKILLS

Languages

C#, C, C++, Python, SQL, x86/x86-64 assembly

Algorithms

Graphs, SCC, max-flow, affine reasoning, conservative program analysis

Communication languages

Russian — native; English — technical documentation and written communication, with no claimed CEFR certificate.

Compiler/runtime

Lexer, parser, AST, bytecode, AIR/IR, interpreter, CIL, DynamicMethod, SSA design

Quality

Regression tests, reference oracles, semantic parity checks, sanitizers, CMake/test runners, benchmark validation

RECOGNITION

2026 · MCST

LLVM-track technical selection

Technical selection result: 1st among 49 submissions, 5.0/5.0 and 104/104 tests on a memory-dependence analysis task.

[Solution](#) · [Internship program](#)

2025-2026

Olympiad and research-project results

Prize-winner of the HSE “Vysshaya Proba” Olympiad in industrial and competitive programming; highest-tier diplomas in the MEPhI Junior Research Competition in 2025 and 2026; winner of the Computer Science track in the HSE “Vysshiiy Pilotazh” Research Competition.

[MEPhI Junior 2025](#) · [MEPhI Junior 2026](#)

EDUCATION & AVAILABILITY

2026-2030

HSE University · Software Engineering

Faculty of Computer Science · bachelor's programme 09.03.04
Moscow · officially admitted; full-time studies begin Sep 2026.

From Sep 2026

Internship or part-time role

Moscow / remote